SMART INDIA HACKATHON 2026 | LOGISTICS SECTOR

CargoChain

Unified Multimodal Freight Visibility & Coordination Platform

India's logistics data already exists in ULIP. We are building the proactive intelligence layer that predicts delays and guides operators before they happen, not after.

Team: CargoChain Developers
Problem Statement ID: SIH-1234
Ministry of Ports, Shipping and Waterways

The Problem Statement

High Logistics Cost (~14% GDP)

India's logistics costs stand at ~14% of GDP, compared to 8-9% in developed nations.

This places Indian manufacturing and export sectors at a competitive disadvantage globally.

Siloed Transport Modes

Rail (FOIS), Road (FASTag/Vahan), and Ports (PCS1x) run on fragmented systems.

There is no unified view showing where cargo is during transition legs.

Extreme Handoff bottlenecks

The longest delays happen when cargo transitions between modes (e.g. Road to Rail, Rail to Vessel).

Vehicles wait idle because the next transport leg is not synchronized.

Downstream Cascading Delays

Delays are realized only after they occur, cascading through subsequent supply chain legs.

Lack of predictive analysis results in reactive adjustments, which multiply expenses.

Existing Landscape & The Crucial Gap

Existing Landscape (Building on It)

• ULIP Integration: The government's Unified Logistics Interface Platform integrates 43 systems, 11 ministries, and 129 APIs.

• Mode Data Available: Systems like FOIS (Rail), PCS1x (Port), and FASTag/Vahan (Road) are already operational.

• Reactive Tracking Solutions: Commercial companies offer GPS vehicle tracking. They show 'where the cargo is' in real-time.

• Focus: Pure data connectivity and post-facto status visualization.

The Gap: Proactive Optimization

• No Proactive Prediction: Current platforms only track. They fail to predict where the cargo will get stuck.

• No Coordinated Instructions: Operators receive data but no actionable decisions (e.g. 'hold truck departure').

• Lack of Dynamic Handoffs: If a train is delayed, port cargo yards and shipping vessels aren't dynamically scheduled, leading to heavy demurrage.

• OUR UNIQUE ANGLE: An Intelligence Layer on top of ULIP that predicts bottleneck delays and tells operators exactly what to do.

Proposed Solution & Unique Value Proposition

1. Ingestion Layer

- Reads mock and real ULIP, FOIS, FASTag, and PCS1x endpoints.
- Aggregates modal records into a single unified JSON state timeline.
- Tracks planned vs actual transitions in real-time.

2. ML Prediction Engine

- Analyzes historical leg transit and port queue patterns.
- Generates handoff-leg delay risk scores (e.g., 72% chance of 4+ hour port wait).
- Flags bottlenecks before they happen.

3. Coordination Alerts (USP)

- Converts risk forecasts into concrete actions.
- Tells operators: 'Hold truck departure 90 mins to avoid port queues.'
- Saves demurrage, waiting times, and transport emission waste.

System Architecture & End-to-End Flow

1. Data Ingest

ULIP API Gateways

FOIS (Rail)

FASTag (Road)

PCS1x (Port)

2. Unified Core

Express Backend

Dynamic JSON DB

Realtime WebSockets

Shipment Timelines

3. ML Engine

Delay Prediction

Leg transit model

Handoff risk scoring

Bottleneck classifier

4. Trust Ledger

Custom Blockchain

SHA-256 blocks

Tamper-proof logs

Verified handovers

Hero Feature Scenario: Ludhiana to Mundra Port

PROACTIVE DISPATCH HOLD ALERT: [Ludhiana Siding -> Mundra Port]

Action Required: Hold Truck departure from Ludhiana Siding by 90 minutes. Reason: ML model predicts a 6-hour delay in Vessel-Ready status at Mundra Port due to incoming container queue. Holding prevents truck fuel wastage and port yard congestion.

Step 1: Road Origin

- Truck loads in Ludhiana.
- System polls road GPS & rail siding queue.
- Detects rail rake allotment lags.

Step 2: Predictive Trigger

- Predicts a 72% chance of 4+ hour delay at Mundra Port.
- Decision engine triggers dispatch hold alert.
- Operator notifies driver instantly.

Step 3: Coordinated Arrival

- Truck arrives exactly when train is ready.
- Seamless handover to rail rake.
- Saves demurrage and waiting times.

Technical Stack (Reflecting CargoChain Codebase)

Layer	Technology Used	Role in CargoChain Platform
Frontend Application	React 19, Leaflet, Recharts, Tailwind CSS	Interactive dashboard displaying real-time shipment map paths, multimodal timeline transitions, and bottleneck charts.
Backend & Live Server	Node.js, Express, Socket.io-client	Real-time event synchronization (WebSockets) sending delay risk changes and coordination alerts dynamically to operators.
Security Ledger	Custom Blockchain (SHA-256 blocks in JS)	Provides a transparent, tamper-proof audit trail for cargo handovers (Origin -> Road -> Rail -> Port -> Destination).
Prediction Analytics	Python, Scikit-learn, XGBoost	Analyzes historical port dwell time and rail transit delays to compute risk scores for downstream transit legs.

Development Roadmap & Future Scope

Phase 1: MVP Core (Hackathon Priority)

- Completed unified data schema for Road/Rail/Port modes.
- Real-time map paths & active WebSocket alerts in React.
- Mocking of ULIP API endpoints (FOIS, FASTag, PCS1x) in JSON DB.

Phase 2: Stretch Feature (Ripple Visualizer)

- Modeling downstream impact cascade (Gantt ripple visualizer).
- Predict how a 2-hour rail delay cascades to port warehouse queue.
- Automate alternative carrier re-allocation routing.

Production Rollout Blueprint

- Replace mock APIs with production ULIP Gateway (NICDC Sandbox).
- Authenticate FOIS & FASTag credentials for registered transport fleet.
- Enterprise SaaS model with automated driver WhatsApp alerts.

Platform Impact, Feasibility & Vision

Logistics GDP Cost Reduction

- India targets logistics cost reduction from 14% to ~9% GDP.
- CargoChain directly addresses transition-leg inefficiencies.
- Potential to reduce port dwell times by 20-30%.

Asset Utilization

- Eliminates truck idle waiting time at ports/sidings.
- Lowers demurrage fees and container retention charges.
- Improves carrier operational turnover speed.

Feasible Implementation

- Builds ON TOP of ULIP rather than replacing it.
- Blockchain verification blocks disputes in multimodal handovers.
- High integration feasibility with existing government systems.

India's logistics data is connected. We supply the intelligence layer that makes it actionable.